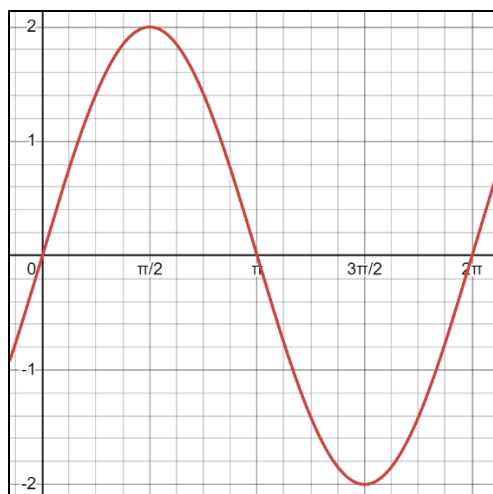
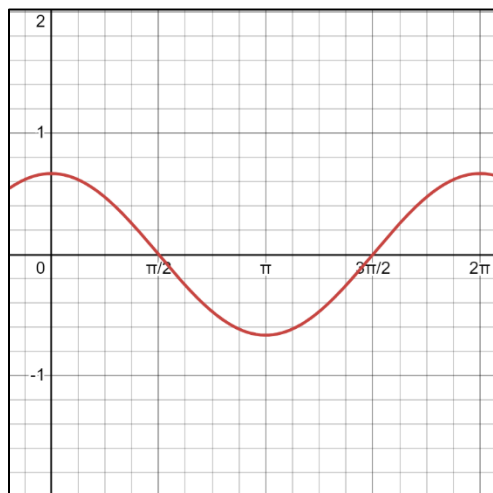


Section 6.1 Exercise Answers

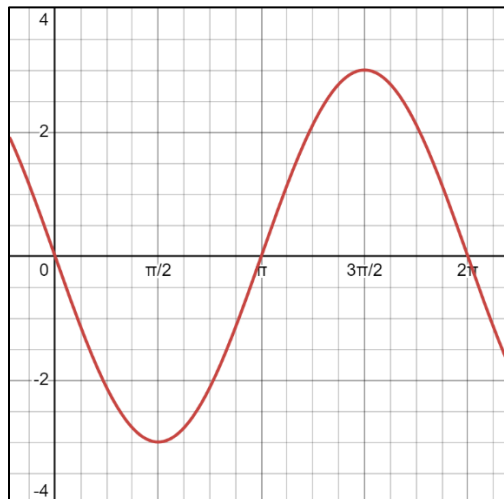
1. Amplitude: 2, Period: 2π , Midline: $y = 0$, Maximum: $(\frac{\pi}{2}, 2)$, Minimum: $(\frac{3\pi}{2}, -2)$



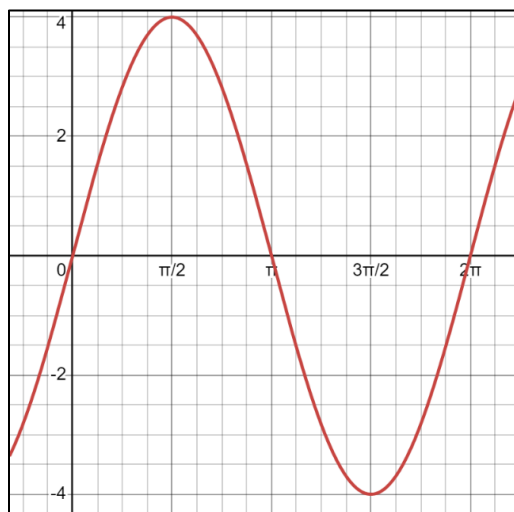
2. Amplitude: $\frac{2}{3}$, Period: 2π , Midline: $y = 0$, Maximum: $(0, \frac{2}{3})$, Minimum: $(\pi, -\frac{2}{3})$



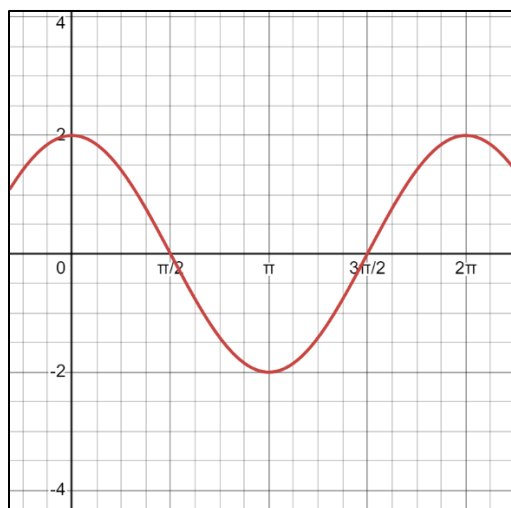
3. Amplitude: 3, Period: 2π , Midline: $y = 0$, Maximum: $(\frac{3\pi}{2}, 3)$, Minimum: $(\frac{\pi}{2}, -3)$



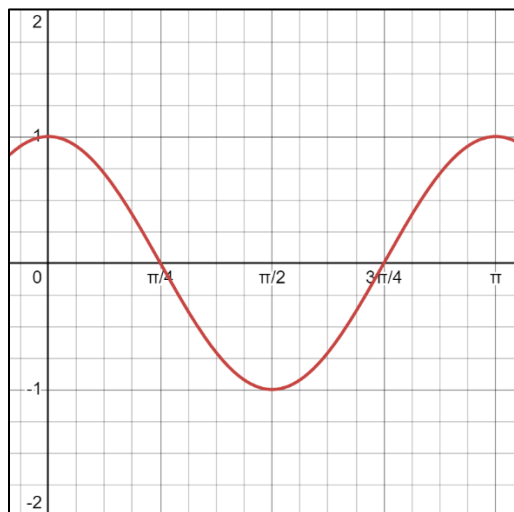
4. Amplitude: 4, Period: 2π , Midline: $y = 0$, Maximum: $(\frac{\pi}{2}, 4)$, Minimum: $(\frac{3\pi}{2}, -4)$



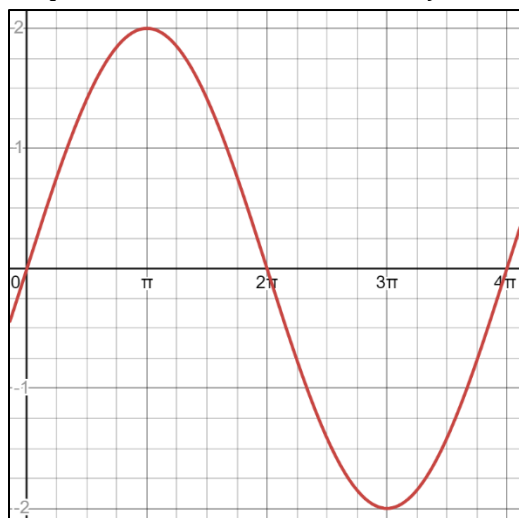
5. Amplitude: 2, Period: 2π , Midline: $y = 0$, Maximum: $(0, 2)$, Minimum: $(\pi, -2)$



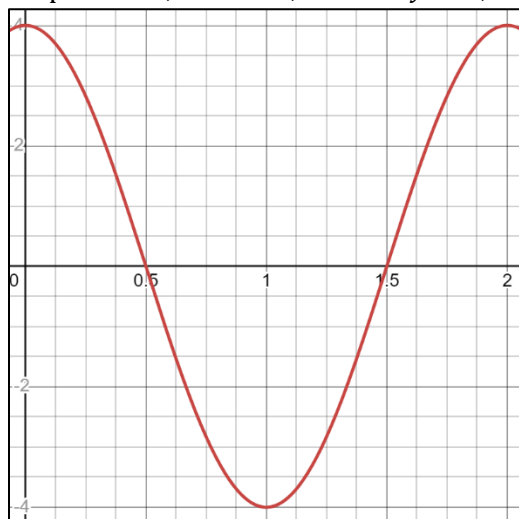
6. Amplitude: 1, Period: π , Midline: $y = 0$, Maximum: $(0, 1)$, Minimum: $(\frac{\pi}{2}, -1)$



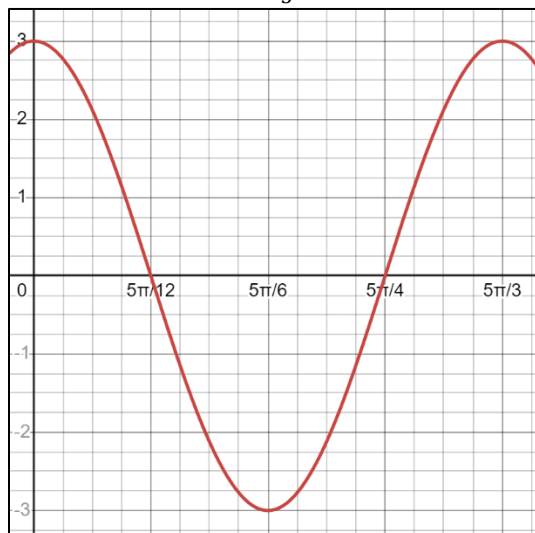
7. Amplitude: 2, Period: 4π , Midline: $y = 0$, Maximum: $(\pi, 2)$, Minimum: $(3\pi, -2)$



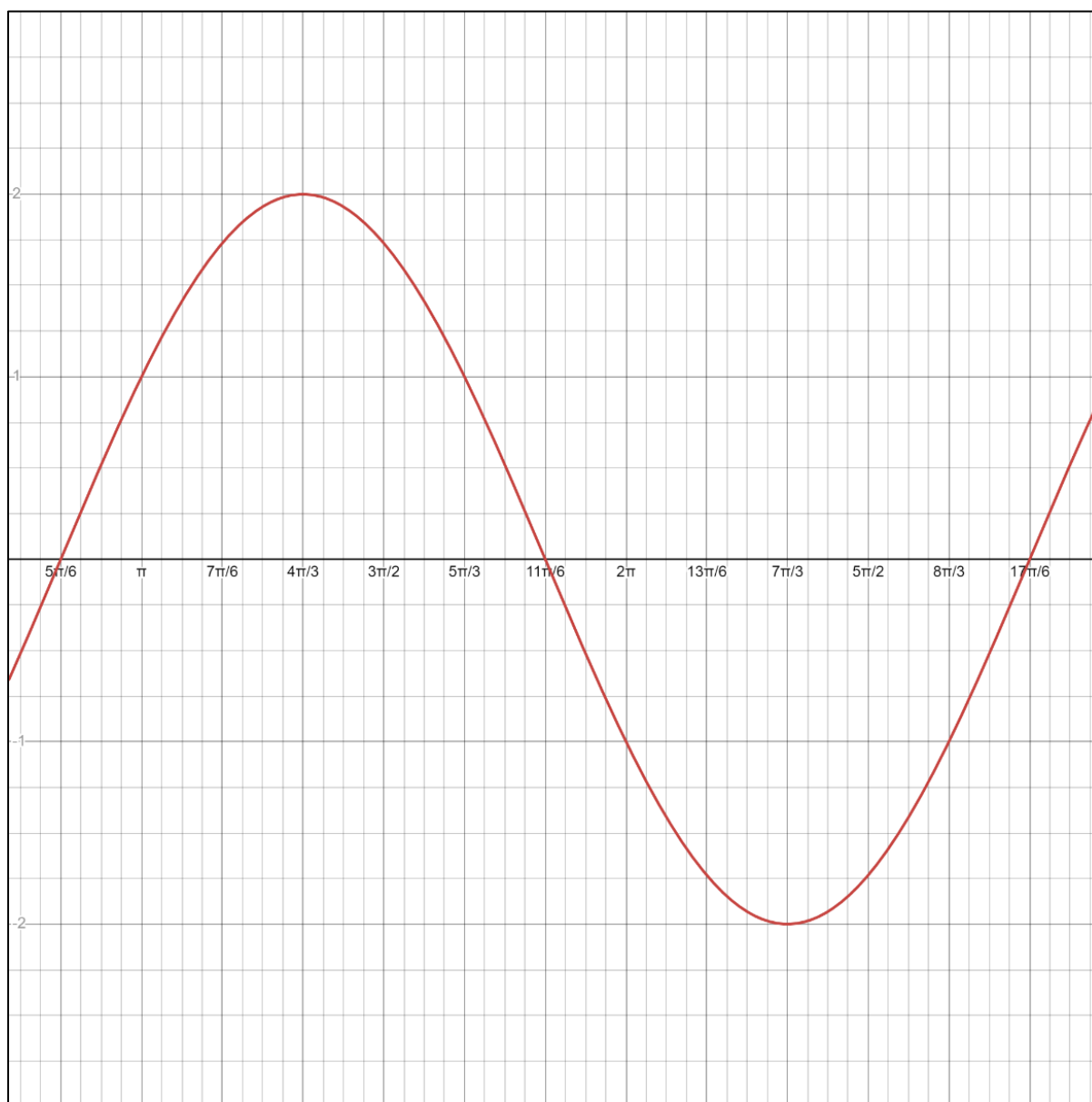
8. Amplitude: 4, Period: 2, Midline: $y = 0$, Maximum: $(0, 4)$, Minimum: $(1, -4)$



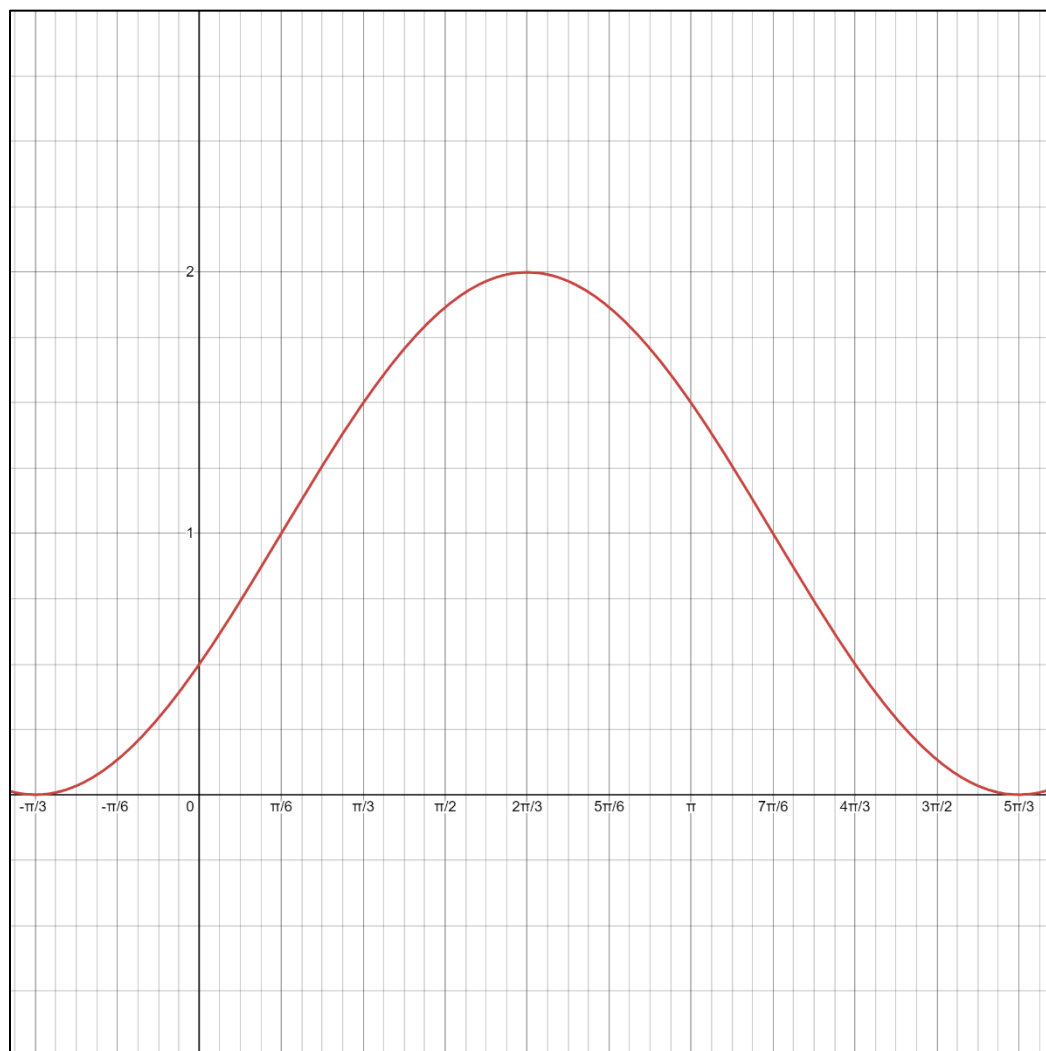
9. Amplitude: 3, Period: $\frac{5\pi}{3}$, Midline: $y = 0$, Maximum: $(0, 3)$, Minimum: $(\frac{5\pi}{6}, -3)$



10. Amplitude: 2, Period: 2π , Midline: $y = 0$, Maximum: $\left(\frac{4\pi}{3}, 2\right)$, Minimum: $\left(\frac{7\pi}{3}, -2\right)$, Phase Shift: Right $\frac{5\pi}{6}$

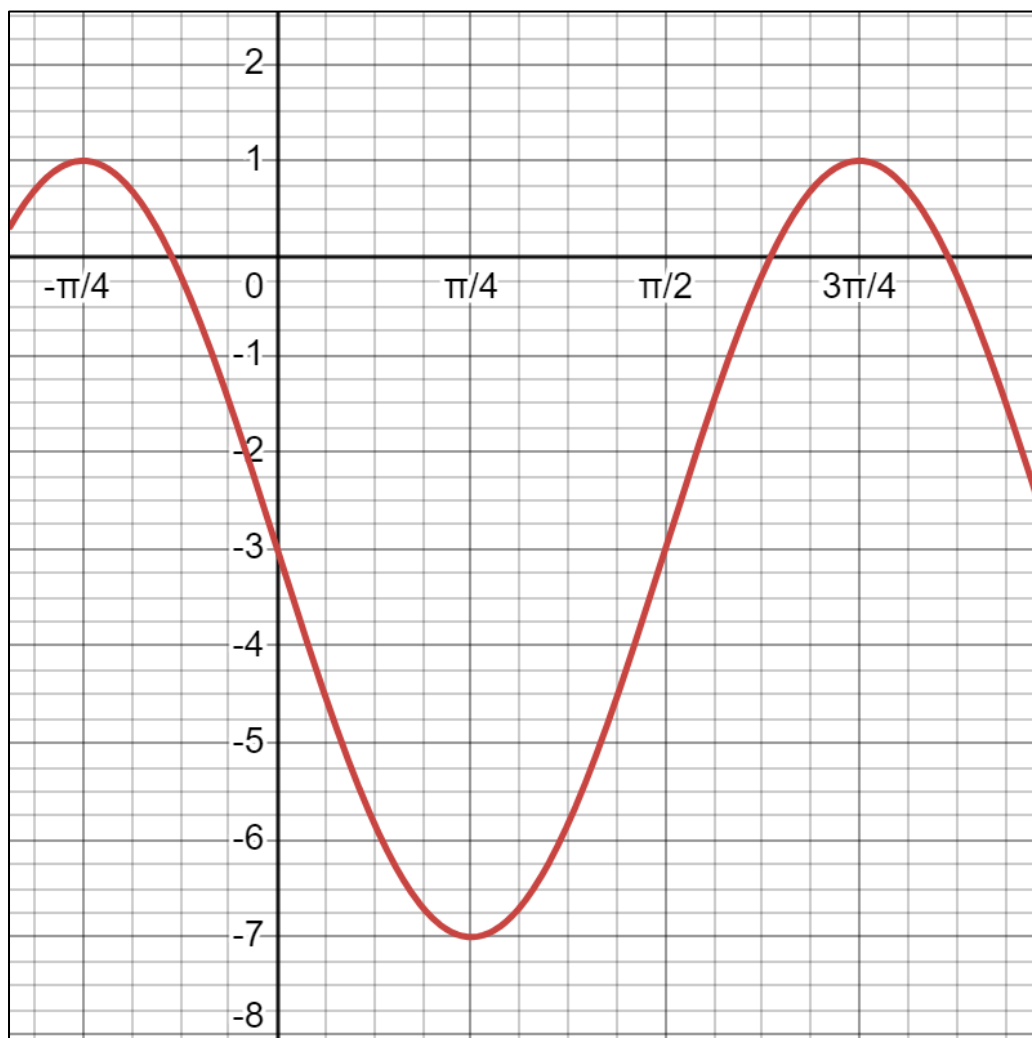


11. Amplitude: 1, Period: 2π , Midline: $y = 1$, Maximum: $(\frac{2\pi}{3}, 2)$, Minimum: $(\frac{5\pi}{3}, 0)$, Phase Shift: Left $\frac{\pi}{3}$

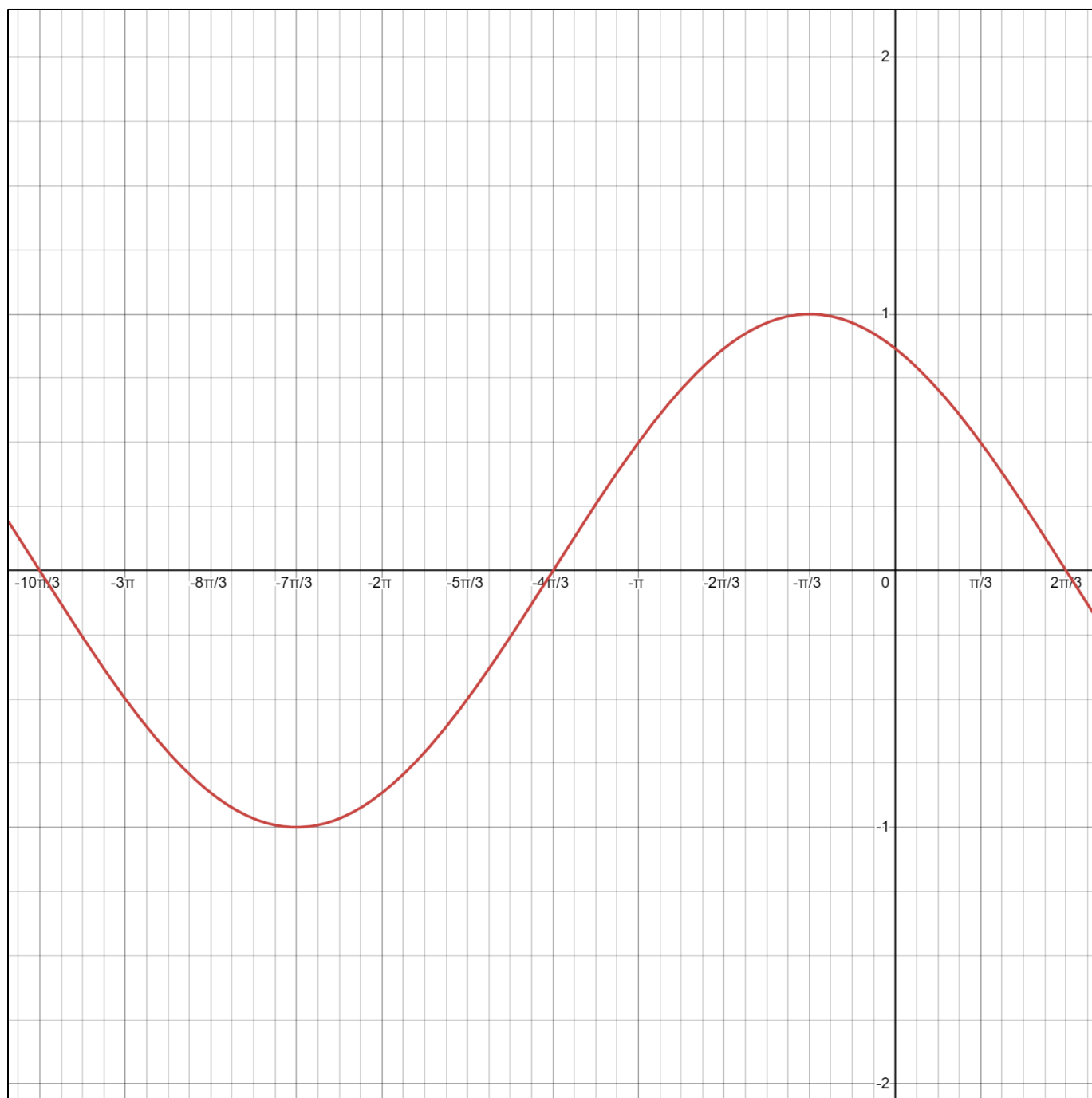


12. Amplitude: 4, Period: π , Midline: $y = -3$,

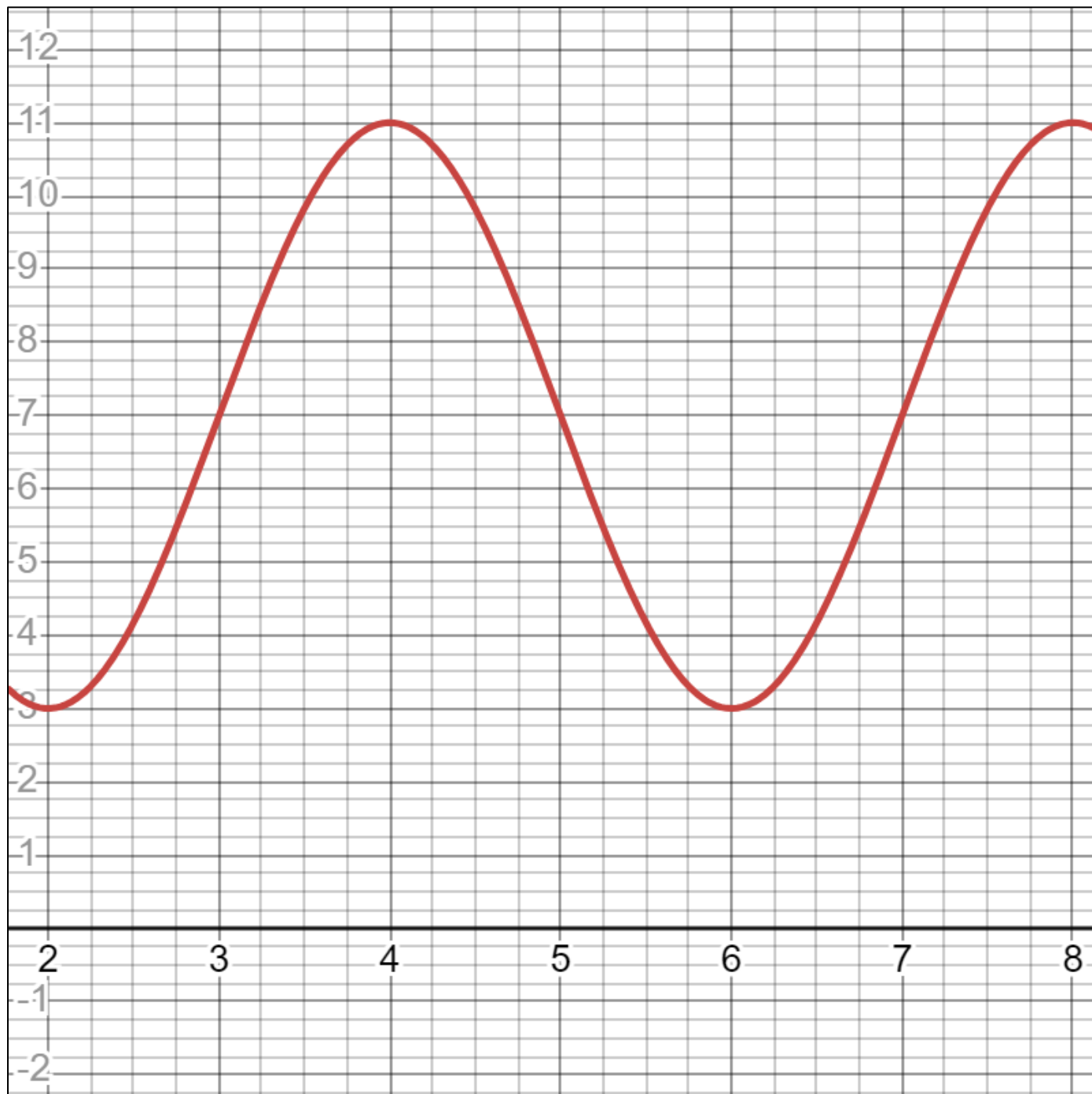
Maximum: $\left(\frac{2\pi}{3}, 2\right)$, Minimum: $\left(\frac{5\pi}{3}, 0\right)$, Phase Shift: Left $\frac{\pi}{4}$, Vertical Translation: Down 3



13. Amplitude: 1, Period: 4π , Midline: $y = 0$, Maximum: $\left(-\frac{\pi}{3}, 1\right)$, Minimum: $\left(-\frac{7\pi}{3}, -1\right)$, Phase Shift: Left $\frac{10\pi}{3}$



14. Amplitude: 4, Period: 4, Midline: $y = 7$,
Maximum: (4,11), Minimum: (6,3), Phase Shift: Right 3, Vertical Translation: Up 7



15. $y = 2 \sin\left(\frac{\pi}{2}x\right) - 3$

16. $y = 3 \cos(2x) - 1$

17. $y = -2 \cos\left(\frac{2\pi}{5}x\right) + 3$

18. $y = 4 \sin\left(\frac{\pi}{5}(x + 1)\right)$